Discrete Math W6

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October 27, 2026

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Hello

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Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Where Are We?

- Basic Logic & Proofs
- ⇒ Sets & Relations
- Number Theory
- Algebra
- Logic

3. Sets & Relations

- Introduction
- Sets and Operations on Sets
- Relations
- Equivalence Relations
- Partial Order Relations
- Functions
- Countable and Uncountable Sets

Functions

Definition 3.33

A *function* $f : A \to B$ from a *domain* A to a *codomain* B is a relation from A to B with the special properties (using the relation notation $a\ f\ b$):

1. $\forall a \in A\ \exists b \in B \quad a\ f\ b$ (f is totally defined),
2. $\forall a \in A\ \forall b, b' \in B\ \left(a\ f\ b \wedge a\ f\ b' \to b = b'\right)$ (f is well-defined).

Definition 3.34

The set of all functions $A \to B$ is denoted as B^A.

$\mathbb{N}^{\mathbb{N}}$ are all functions from $\mathbb{N}$ to $\mathbb{N}$

Partial functions

Definition 3.35

A *partial function* $A \to B$ is a relation from A to B such that condition 2. above holds.

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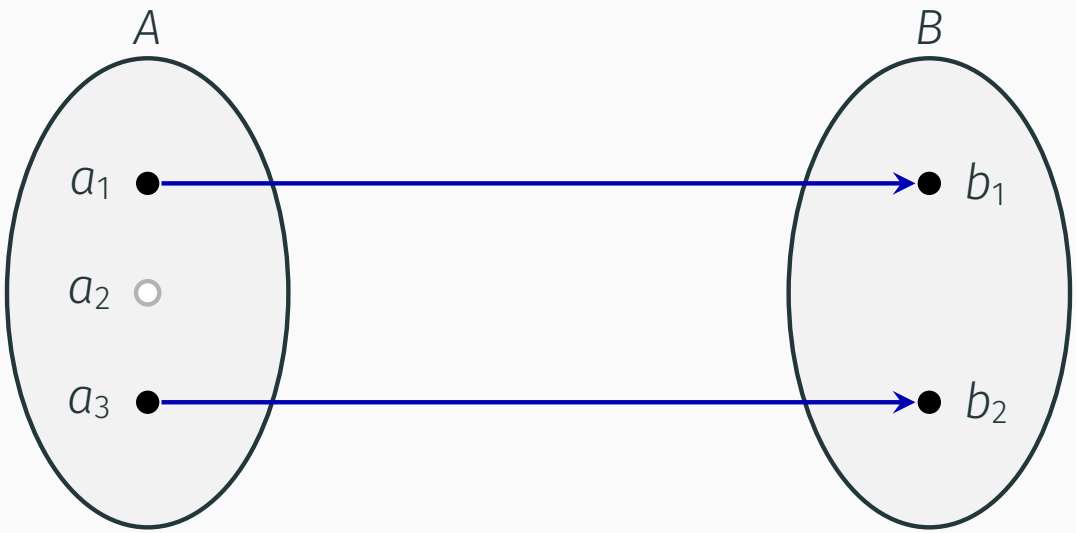


Image / Range / Preimage

Definition 3.36

For a function $f : A \to B$ and a subset S of A, the *image* of S under f, denoted $f(S)$, is the set

$$f(S) \stackrel{\text{def}}{=} \{f(a) \mid a \in S\}.$$

Definition 3.37

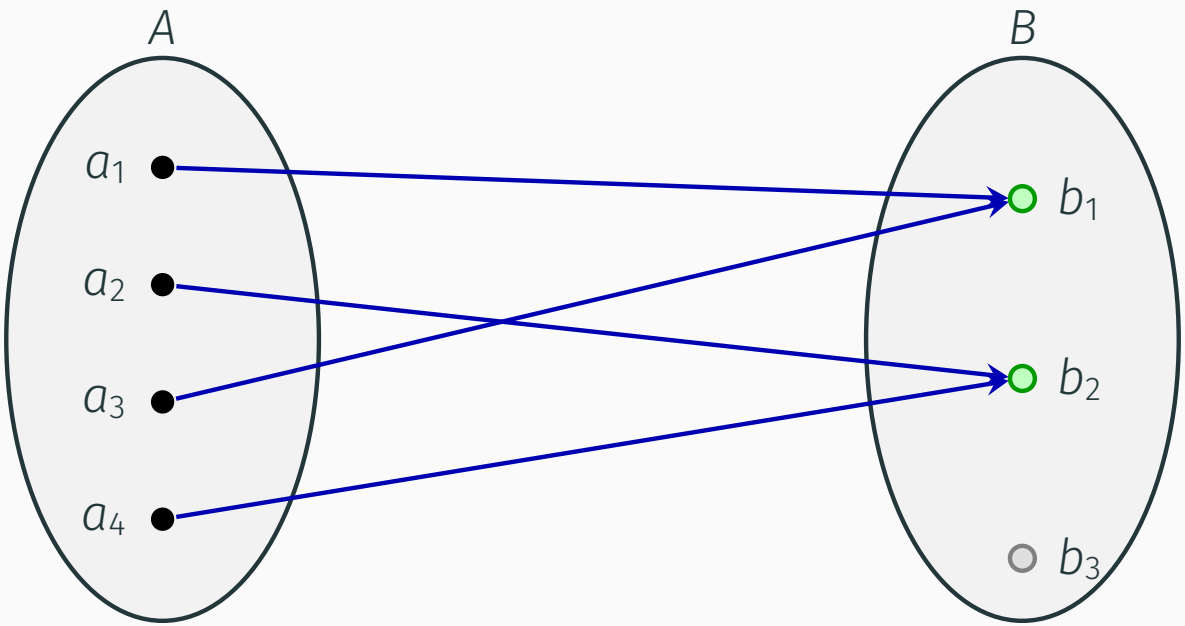
The subset $f(A)$ of B is called the *image* (or *range*) of f and is also denoted $\mathsf{Im}(f)$.

Definition 3.38

For a subset T of B, the *preimage* of T, denoted $f^{-1}(T)$, is the set of values in A that map into T:

$$f^{-1}(T) \stackrel{\text{def}}{=} \{a \in A \mid f(a) \in T\}.$$

Image, Range and Preimage



Range / Image of A: $\mathrm{Im}(f) = f(A) = \{b_1, b_2\} \subsetneq B$ (green)

Image, Range and Preimage

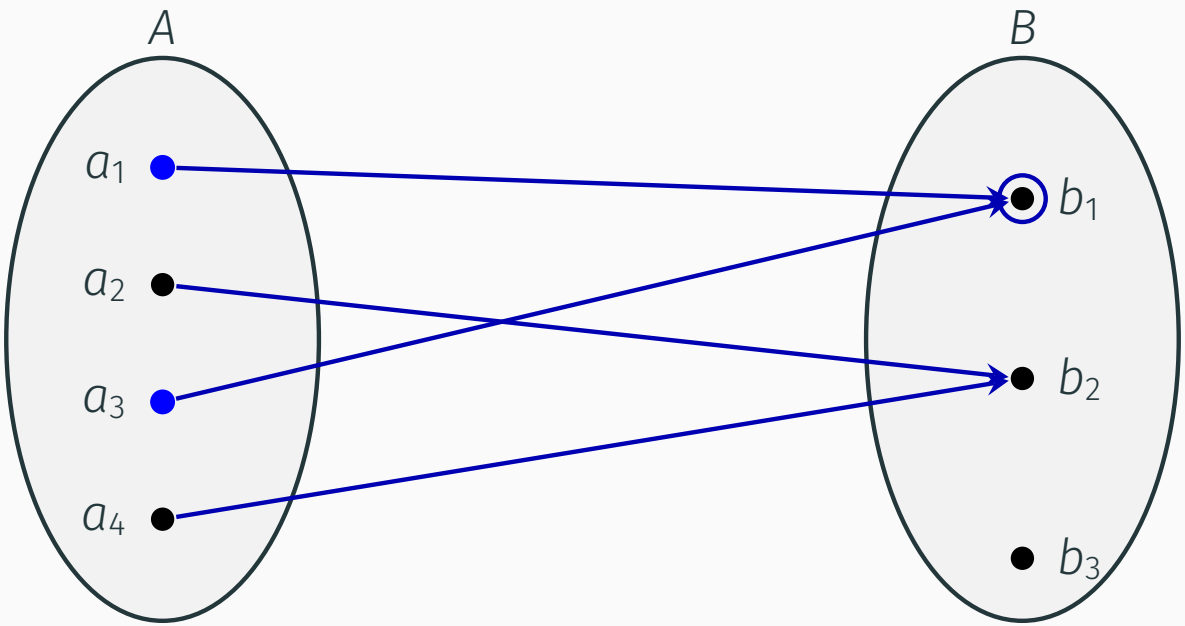
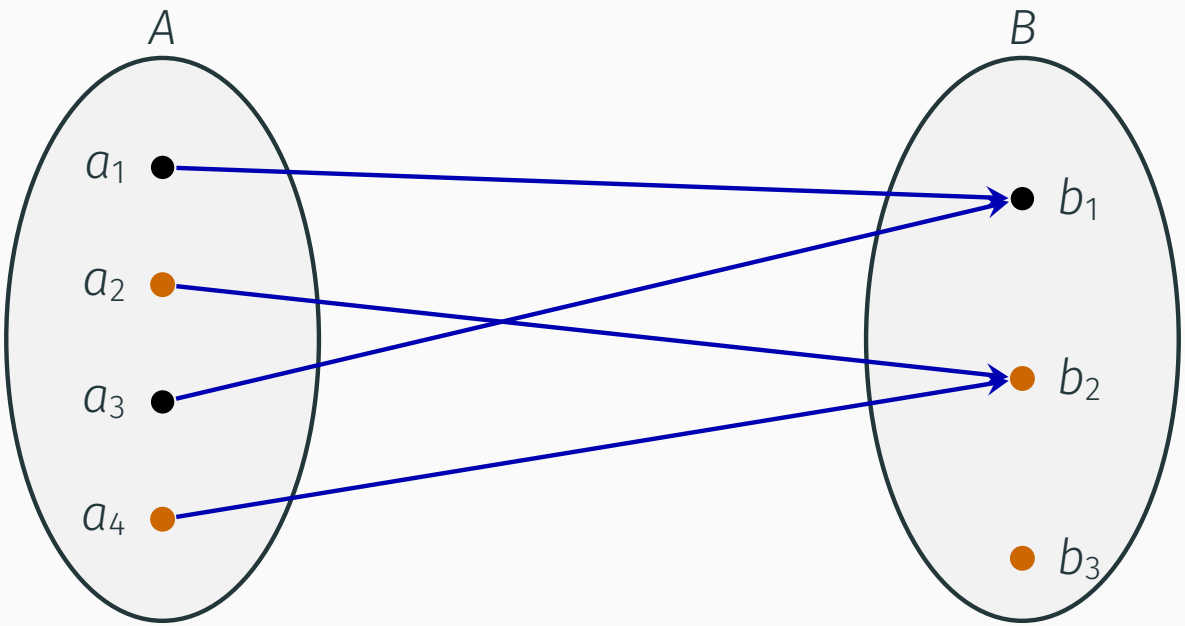


Image of $S = \{a_1, a_3\}$: $f(S) = \{b_1\}$ (blue)

Image, Range and Preimage



Preimage of $T = \{b_2, b_3\}$: $f^{-1}(T) = \{a_2, a_4\}$ (orange)

Injective etc

Definition 3.39

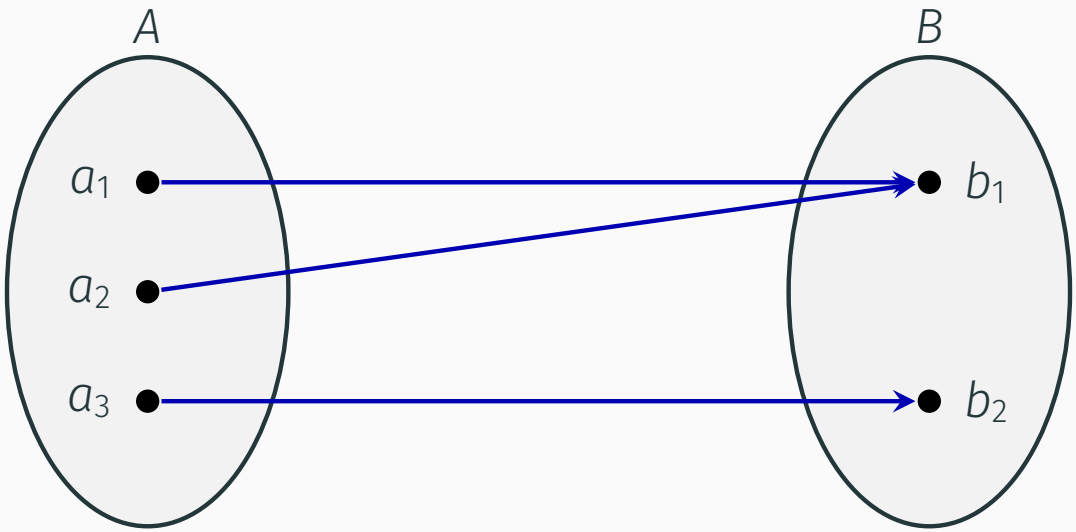
A function $f : A \to B$ is called

1. *injective* (or *one-to-one* or an *injection*) if for $a \neq a'$ we have $f(a) \neq f(a')$, i.e., no two distinct values are mapped to the same function value (there are no "collisions").
2. *surjective* (or *onto*) if $f(A) = B$, i.e., if for every $b \in B$, $b = f(a)$ for some $a \in A$ (every value in the codomain is taken on for some argument).
3. *bijective* (or a *bijection*) if it is both injective and surjective.

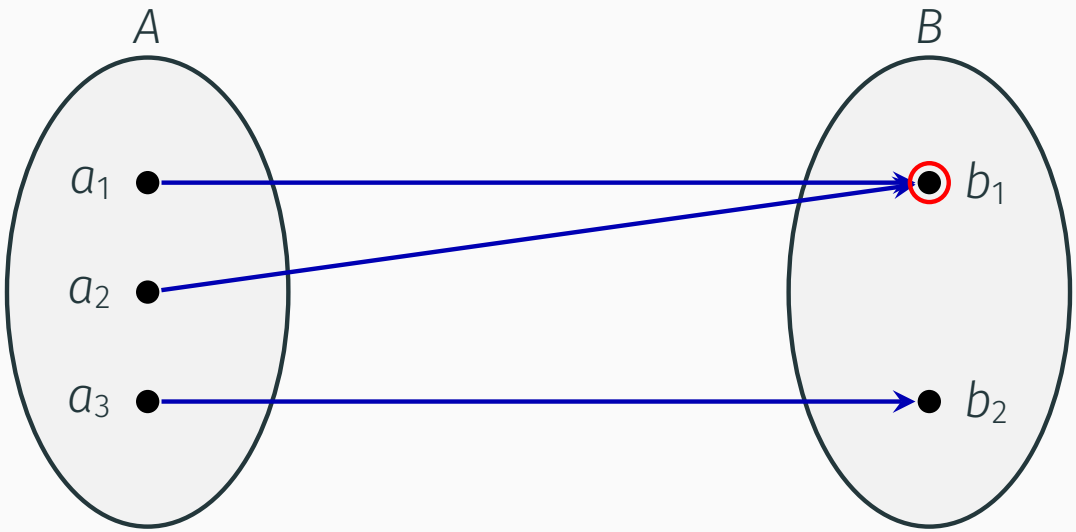
Definition 3.40

For a bijective function f, the *inverse* (as a relation, see Definition 3.11) is called the *inverse function* of f, usually denoted as f^{-1}.

Example



Example



$a_1 \neq a_2$ but $f(a_1) = f(a_2) = b_1$ not injective

Function composition

Definition 3.41

The *composition* of a function $f : A \to B$ and a function $g : B \to C$, denoted by $g \circ f$ or simply gf, is defined by $(g \circ f)(a) = g(f(a))$.

Lemma 3.14

Function composition is associative, i.e., $(h \circ g) \circ f = h \circ (g \circ f)$.

Countability of sets

Definition 3.42

(i) Two sets A and B *equinumerous*, denoted $A \sim B$, if there exists a bijection $A \to B$.

(ii) The set B *dominates* the set A, denoted $A \preceq B$, if $A \sim C$ for some subset $C \subseteq B$ or, equivalently, if there exists an injective function $A \to B$.

(iii) A set A is called *countable* if $A \preceq \mathbb{N}$, and *uncountable* otherwise.

Between finite and countably infinite

Lemma 3.15

(i) The relation $\sim$ is an equivalence relation.

(ii) The relation $\preceq$ is transitive: $A \preceq B \wedge B \preceq C \implies A \preceq C$.

(iii) $A \subseteq B \implies A \preceq B$.

Theorem 3.16

$A \preceq B \wedge B \preceq A \implies A \sim B$.

Theorem 3.17

A set A is countable if and only if it is finite or if $A \sim \mathbb{N}$.

Important countable sets

Theorem 3.18

The set $\{0,1\}^* \stackrel{\text{def}}{=} \{\epsilon, 0, 1, 00, 01, 10, 11, 000, 001, \ldots\}$ of finite binary sequences is countable.

Theorem 3.19

The set $\mathbb{N} \times \mathbb{N}$ $(= \mathbb{N}^2)$ of ordered pairs of natural numbers is countable.

Corollary 3.20

The Cartesian product $A \times B$ of two countable sets A and B is countable, i.e., $A \preceq \mathbb{N} \wedge B \preceq \mathbb{N} \implies A \times B \preceq \mathbb{N}$.

Corollary 3.21

The rational numbers $\mathbb{Q}$ are countable.

Theorem 3.22

Let A and A_i for $i \in \mathbb{N}$ be countable sets.

(i) For any $n \in \mathbb{N}$, the set A^n of n-tuples over A is countable.

(ii) The union $\bigcup_{i \in \mathbb{N}} A_i$ of a countable list $A_0, A_1, A_2, \ldots$ of countable sets is countable.

(iii) The set A^* of finite sequences of elements from A is countable.

Uncountability of $\{0,1\}^\infty$

Definition 3.43

Let $\{0,1\}^\infty$ denote the set of semi-infinite binary sequences or, equivalently, the set of functions $\mathbb{N} \to \{0,1\}$.

Theorem 3.23

The set $\{0,1\}^\infty$ is uncountable.

Cantor's Diagonalization Method

$f(i)$	0	1	2	3	4	5	...
$f(0)$	**0**	0	1	1	0	1	...
$f(1)$	1	**1**	0	0	1	0	...
$f(2)$	0	1	**1**	0	0	1	...
$f(3)$	1	0	0	**1**	1	0	...
$f(4)$	0	1	0	1	**0**	0	...
$f(5)$	1	1	0	0	1	**1**	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
α	**1**	**0**	**0**	**0**	**1**	**0**	...

Construct a new sequence α by **flipping** each diagonal bit: $\alpha_i = 1 - f(i)_i$.

Conclusion: α differs from every $f(i)$ at position i, so α cannot be in the list.

Thus, $\{0,1\}^{\mathbb{N}}$ cannot be enumerated.

Uncomputable functions

Definition 3.44

A function $f : \mathbb{N} \to \{0, 1\}$ is called *computable* if there is a program that, for every $n \in \mathbb{N}$, when given n as input, outputs $f(n)$.

Corollary 3.24

There are uncomputable functions $\mathbb{N} \to \{0, 1\}$.

Techniques for proving (Un)countability

For proving that a given set A is countable there are three main high-level strategies

1. Find an explicit bijection with $\mathbb{N}$
2. Express A as the countable union of countable sets
3. Find an injection from A to a known countable set B

For proving uncountability we have

1. Find an injection from a known uncountable set like $\{0,1\}^\infty$ to A
2. Use Cantor's diagonalization argument.

Standard proof structure

First you should get a good understanding of your given set. Try to find similarities to other sets you already know. Then a standard pattern for proving that a set A is uncountable is as follows

1. Find an injective function $f : \{0,1\}^\infty \to A$. You need to encode all information of the bitstring b in the image $f(b)$
2. **Prove that f is well-defined and totally defined**
3. **Prove that f actually maps to your set A**
4. Prove that f is injective
5. Conclude that $\{0,1\}^\infty \preceq A$ and hence A is uncountable.

For the last step we need a bit more detail since $\{0,1\}^\infty \preceq A$ does not directly show that A is uncountable.

So assume that A is countable. So $A \preceq \mathbb{N}$ and by transitivity $\{0,1\}^\infty \preceq \mathbb{N}$, which is a contradiction. Hence A is uncountable.

Some tricks

These are some tricks that can be useful for doing such proofs

1. Find an uncountable set B such that $A \subseteq B$ and prove that $B \setminus A$ is countable. (We will see this later)
2. Use the ordering of a given set find an injective function
3. Split the set using set operations and argue on each one seperately

Exercises

Show that if $A \subseteq B$ and $f : B \to A$ is an injection, then $|A| = |B|$.

Show that if $A \subseteq B$ and $f : B \to A$ is an injection, then $|A| = |B|$.

Proof.

Let $g : A \to B$ be defined as $g(x) = x$. Since $A \subseteq B$, this implies that g maps to B and is hence well defined. Clearly g is injective.

Now there exist two injections $f : B \to A$ and $g : A \to B$. By Theorem 3.16 we conclude that $A \sim B$ □

Let A be a set and let $id_A : A \to A$ the identity function. Prove that $f : A \to A$ is bijective if and only if there exists a function $g : A \to A$ such that

$$g \circ f = id_A \quad \text{and} \quad f \circ g = id_A$$

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Proof.

$\Longrightarrow$: Assume that f is bijective. Then f is both injective and surjective. By definition this means that for every $y \in A$ there exists a unique $x \in A$ such that $f(x) = y$.

We construct a function $g : A \to A$ by defining $g(y) = x$, where x is the unique element satisfying $f(x) = y$.

Now we check the two conditions.

- Left inverse: For any $x \in A$ let $y = f(x)$ then $(g \circ f)(x) = g(f(x)) = g(y) = x = id_A$
- Right inverse: For any $y \in A$ let $x = g(y)$ then $(f \circ g)(y) = f(g(y)) = f(x) = y = id_A$

□

Proof.

$\Longleftarrow$: Assume there exists a function $g : A \to A$ such that

$$g \circ f = id_A \quad \text{and} \quad f \circ g = id_A$$

Let $x_1, x_2 \in A$ such that

$$\begin{aligned} f(x_1) &= f(x_2) \\ g(f(x_1)) &= g(f(x_2)) \\ (g \circ f)(x_1) &= (g \circ f)(x_2) \\ x_1 &= x_2 \end{aligned}$$

So f is injective

Let $y \in A$ be arbitrary. Let $x = g(y)$. Then we get

$$f(x) = f(g(y)) = (f \circ g)(y) = id_A(y) = y$$

Since we found an $x \in A$ for every $y \in A$, f is surjective.

□

Irrational numbers

Prove or disprove. The irrational numbers are countable.

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Proof.

We prove that the irrational numbers are uncountable.

We define $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ as the set of irrational numbers. Then notice that we can write $\mathbb{R} = \mathbb{I} \cup \mathbb{Q}$, where $\mathbb{Q}$ is countable.

Now assume for sake of contradiction that $\mathbb{I}$ is countable. Then $\mathbb{R}$ is the union of countable sets and hence countable, which is a contradiction.

So we conclude that $\mathbb{I}$ is uncountable. □

Exam HS 22

($\star\,\star$) Consider the set

$$S = \{A \subseteq \mathbb{N} \mid A \text{ or } \mathbb{N} \setminus A \text{ is finite}\}.$$

Prove or disprove that the set S is countable. *(5 Points)*

Proof.

Notice that S can be written as the union of two sets F, G such that

$$\begin{aligned} S &= F \cup G \\ &= \{A \subseteq \mathbb{N} \mid A \text{ is finite}\} \cup \{A \subseteq \mathbb{N} \mid \mathbb{N} \setminus A \text{ is finite}\} \end{aligned}$$

Let F_n be the set of subsets of $\mathbb{N}$ with exactly n elements. By the well ordering of $\mathbb{N}$ under $<$ we can order each subset and represent it uniquely as a strictly increasing tuple of elements $(a_1, \ldots, a_n) \in \mathbb{N}^n$. This defines an injective mapping from F_n to $\mathbb{N}^n$.

By Th. 3.22 $\mathbb{N}^n$ is countable, so F_n is countable for any $n \geq 0$.

We can write $F = \bigcup_{n \in \mathbb{N}} F_n$ as the countable union of countable sets, so we get that F is countable.

Define a function $\psi : G \to F$ by $\psi(A) = \mathbb{N} \setminus A$.

If $A \in G$ then by definition $\mathbb{N} \setminus A$ is finite. So $\psi(A) = \mathbb{N} \setminus A \in F$ and ψ is well defined.

□

Proof.

If $\psi(A_1) = \psi(A_2)$ then $\mathbb{N} \setminus A_1 = \mathbb{N} \setminus A_2$, which implies that $A_1 = A_2$ and ψ is injective.

Let $B \in F$ be arbitrary. Let $A = \mathbb{N} \setminus B$. Then $\psi(A) = \mathbb{N} \setminus (\mathbb{N} \setminus B) = B$, since $B \subseteq \mathbb{N}$.

So ψ is surjective and hence a bijection.

We get that $G \sim F$ and since F is countable, G is also countable.

As S is the union of countable sets, we conclude that S is countable. □

Prove or disprove that the following set is uncountable

$$A = \{f \in \{0,1\}^{\mathbb{N}} \mid f(k) = 0 \text{ if } k \equiv_2 0\}$$

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$$A = \{f \in \{0,1\}^{\mathbb{N}} \mid f(k) = 0 \text{ if } k \equiv_2 0\}$$

Proof.

We prove that A is uncountable by constructing an injection $f : \{0,1\}^{\infty} \to A$. Let $b \in \{0,1\}^{\infty}$ be any bitstring and b_i its i-th bit.

We define f as $b \mapsto g_b$ where we map every bitstring to a corresponding function g_b

$$g_b(k) = \begin{cases} 0 & \text{if } k \equiv_2 0 \\ b_{\lfloor k/2 \rfloor} & \text{if } k \equiv_2 1 \end{cases}$$

Let $b \in \{0,1\}^{\infty}$ be arbitrary. Then by definition we have $f(b) \in A$, since for every even k we get $f(b)(k) = 0$. Further we map each bitstring to exactly one function g_b, so f is well-defined and so a function

Now we prove that f is injective. Let $b, b' \in \{0,1\}^{\infty}$ such that $b \neq b'$. Then there must exist some index $j \in \mathbb{N}$ such that $b_j \neq b'_j$

Proof.

By definition of f we get that

$$f(b)(2 \cdot j + 1) = g_b(2 \cdot j + 1) = b_j$$
$$f(b')(2 \cdot j + 1) = g_{b'}(2 \cdot j + 1) = b'_j$$

This implies that $f(b) \neq f(b')$ and so f is injective.

Now we have $\{0,1\}^\infty \preceq A$. Now assume for sake of contradiction the A is countable. Then $A \preceq \mathbb{N}$ and by transitivity $\{0,1\}^\infty \preceq \mathbb{N}$. But this implies that $\{0,1\}^\infty$ is countable, which is a contradiction.

Hence we conclude that A is uncountable. □

Exam HS 23

($\star\,\star$) Consider the set of real numbers that are not roots of polynomials with rational coefficients:

$$A = \{\alpha \in \mathbb{R} \mid p(\alpha) \neq 0 \text{ for all non-zero polynomials } p(x) \in \mathbb{Q}[x]\}.$$

Prove or disprove that the set A is countable. *Hint*: first prove that $\mathbb{Q}[x]$ is countable. *(11 Points)*

Think of $\mathbb{Q}[x]$ as tuples $(a_1, a_2, \ldots)$ where $a_i \in \mathbb{Q}$

First prove that $\mathbb{Q}_n[x]$ is countable.

Every non-constant polynomial of degree n has at most n roots in $\mathbb{R}$

Proof.

We prove that A is uncountable

First notice that we can write $\mathbb{Q}_n[x]$ as (is isomorphic to)

$$\mathbb{Q} \setminus \{0\} \times \underbrace{\mathbb{Q} \times \cdots \times \mathbb{Q}}_{n \text{ times}}$$

Since Q is countable, we get that the n-fold cartesian product is countable and hence $\mathbb{Q}_n[x]$ is countable for any $n \geq 0$.

Then we define $\mathbb{Q}[x]$ as

$$\mathbb{Q}[x] = \{0\} \cup \bigcup_{n \in \mathbb{N}} \mathbb{Q}_n[x]$$

So $\mathbb{Q}[x]$ is a countable union of countable sets, hence $\mathbb{Q}[x]$ is countable.

□

Proof.

Now consider the set $\mathbb{R} \setminus A$ of all distinct roots for all non-zero polynomials in $\mathbb{Q}[x]$.

Since every non-zero polynomial has at most n roots in $\mathbb{R}$ and $\mathbb{Q}[x]$ is countable, we get that $\mathbb{R} \setminus A$ is a countable union of finite sets. So $\mathbb{R} \setminus A$ is countable

Now assume for sake of contradiction that A is countable. Then we can write

$$\mathbb{R} = \mathbb{R} \setminus A \cup A$$

Since both are countable, this implies that $\mathbb{R}$ is the union of countable sets and therefore countable. This is a contradiction.

So we conclude that A is uncountable. □

Exam HS 24

e) $(\star\,\star)$ Let X be a set and let ρ be a countable relation on X. **Prove or disprove** that the transitive closure ρ^* of ρ is countable. *(9 Points)*

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Proof.

Let $A = \{x \in X \mid \exists y \in X \ \ (x, y) \in \rho \vee (y, x) \in \rho\}$. Since ρ is countable and each pair in ρ can at most appear twice in A, we get that A is countable.

Since A is countable, $A \times A$ is also countable.

Now we show that $\rho^n \subseteq A \times A$ for $n \geq 1$.

For $n = 1$, we get by definition of A that $\rho \subseteq A \times A$.

Now assume that $\rho^n \subseteq A \times A$ for some $n \geq 1$. Let $(x, y) \in \rho^{n+1}$.

Using the definition of relation composition we get

$$(x, y) \in \rho^{n+1} \iff \exists z \in X \ (x, z) \in \rho^n \wedge (z, y) \in \rho$$

□

Proof.

Since $(x, z) \in \rho^n$, the induction hypothesis gives that $x, z \in A$.
Further the definition of A and $(z, y) \in \rho$ imply that $z, y \in A$.

So both x and y belong to A and so $(x, y) \in A \times A$. Thus $\rho^{n+1} \subseteq A \times A$.
By induction the claim holds for all $n \geq 1$.

Hence ρ^n is countable for $n \geq 1$.

Since $\rho^* = \bigcup_{n \in \mathbb{N} \setminus \{0\}} \rho^n$ is a countable union of countable sets, we get that ρ^* is countable.

□

Questions?

Material: `https://diegostrebel.com/teaching`

